INCH-POUND

MIL-L-11647E
26 July 1991
SUPERSEDING
MIL-L-11647D
31 August 1972

MILITARY SPECIFICATION

LEVER, BALK DEPRESSING:

CLAW-BAR TYPE

This specific is approved for use within the USA Belvoir Research Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a claw-bar type, balk depressing lever to be used in the erection of fixed and floating bridges having deck-balk superstructures.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood.

MILITARY

MIL-T-704	-	Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Government drawings. The following Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

TA13219E3956	-	Lever, Balk Depressing: Claw-Bar Type.

(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JB, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING MATERIALS (ASTM)

D 3953 - Strapping, Flat Steel and Seals.
D 4675 - Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

A 5.1 - Covered Carbon Steel Arc Welding Electrodes.
A 5.18 - Carbon Steel Filler Metals for Gas Shielded Arc Welding.
D 1.1 - Structural Welding Code - Steel.

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION (NMFTA)

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314-4677.)

UNIFORM CLASSIFICATION COMMITTEE (UCC)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. The claw-bar type, balk depressing lever consisting of a modified crowbar, hooks, chains, and attachments shall be as shown on Drawing TA13219E3956 and as specified herein.

3.1.1 Drawing. The drawing forming a part of this specification is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2) a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and as shown on the drawing. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The balk depressing lever shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environment to which the balk depressing lever may be exposed.

3.3.1.1 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A 5.1 or A 5.18, classification and size as required.

3.5 Performance. The balk depressing lever shall withstand a load of 450 pounds applied 6 inches from the end of the handle while the lever is engaged at the point without permanent deformation in the handle or point, or failure in the welds.

3.6 Identification marking. The balk depressing lever shall be metal-die stamped as shown on the drawing. Die stamping shall be legible after painting.

3.7 Treatment and painting. The balk depressing lever shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, except that the color number designation shall be 34088 in accordance with FED-STD-595.

3.8 Workmanship.

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3.8.1 Steel fabrication. Steel used in the fabrication of the balk depressing lever shall be free from laminations, kinks, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs and rough edges shall be ground smooth.

3.8.2 Welders and welding.

3.8.2.1 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

ASME Boiler and Pressure Vessel Code, Section IX.
AWS D1.1 - Structural Welding Code - Steel.

In the event of evidence of poor welds, the Government reserves the right to require recertification of any welder or welding operator. The test results shall be made available for review by the contracting officer or the contracting officer's representative.

3.8.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the AWS D1.1 or ASME code. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.9 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the balk depressing lever may be newly fabricated from recovered materials to the maximum extent practicable, provided the balk depressing lever produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the balk depressing lever.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The supplier is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article balk depressing lever shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The balk depressing lever shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

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4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Test. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The balk depressing lever shall be examined as specified herein for the following defects:

Major

101. Nonconformance to the dimensions shown on the drawing (see 3.1.1).
102. Materials not as specified (see 3.3).
103. Workmanship not as specified (see 3.8).
104. Welders and welding operators not certified as specified (see 3.8.2.1).
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal (see 3.8.2.2).
106. Welding not as specified (see 3.8.2.2).

Minor

201. Identification marking incorrect, missing, or illegible (see 3.6).
202. Cleaning, treatment, and painting not as specified (see 3.7).

4.5.2 Performance Test. Apply a load of 450 pounds 6 inches from the end of the handle while the lever is engaged at the point as shown in figure 1. Apply the load for not less than 2 minutes. Any permanent deformation in the handle or point or any failure in the welds shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packaging.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects. Presence of one or more defects shall be cause for rejection.

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107. Containers not as specified for level A or B (see 5.1.1 and 5.1.2).
108. Chains not secured to the bar as specified for level A or B (see 5.1.1 and 5.1.2).
109. Blocking, bracing, and anchoring not applied as specified for level A or B (see 5.1.1 and 5.1.2).
110. Strapping not zinc-coated for level A (see 5.1.1).
111. Packing not as specified for level C (see 5.1.3).
112. Marking illegible, incorrect, or incomplete (see 5.2).

5. PACKAGING

5.1 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. The chains shall be secured to the bars with wire ties. The levers shall be packed together in boxes conforming to PPP-B-601, overseas type, style optional, grade B. Blocking, bracing, and anchoring shall be applied in accordance with MIL-STD-1186. Boxes exceeding 200 pounds shall be modified with skids in accordance with the box specification. Closure of the box shall be as specified by the box specification. Strapping shall be in accordance with ASTM D 3953, type 1 or 2, zinc-coated, size as applicable and ASTM D 4675 and on modified boxes shall be placed parallel with and adjacent to the skids.

5.1.2 Level B. The levers shall be packed as specified in 5.1.1, except boxes shall be domestic type and strapping need not be zinc-coated.

5.1.3 Level C. The levers shall be packed or prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at the lowest ratings of the carrier in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking (see 6.4). Shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The lever is intended for use in depressing the balk to facilitate placement of the pins during the erection of fixed and floating bridges having deck balk superstructure. This item is a component of the following Supply Catalogs (SC) 5420-97-CL-E09, E36, E41 and E45.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

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c. When a first article is required for inspection and approval and numbers of units required (see 3.2, 4.3 and 6.3).
d. Level of packing required (see 5.1).

6.3 First Article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation. Any changes or deviations of production balk depressing levers from the approved first article model during production will be subject to the approval of the contracting officer. Approval of the first article model will not relieve the supplier of his obligation to furnish balk depressing levers conforming to this specification.

6.4 Levels of preservation. Levels of preservation have not been included as the levers do not require preservation. For the purpose of marking, the shipping containers shall be marked A/A, A/B or A/C.

6.5 Changes from the previous issue. Marginal notations are not used in this revision to identify changes which respect to the previous issue due to the extensiveness of the changes.

6.6 Subject term (key word) listing.

Balk depressing lever
Claw-bar type
Deck-balk superstructures

Custodians:
Army - ME

Preparing activity:
Army - ME

Review activities:
Navy - MC
DLA - CS

Project 5420-0169

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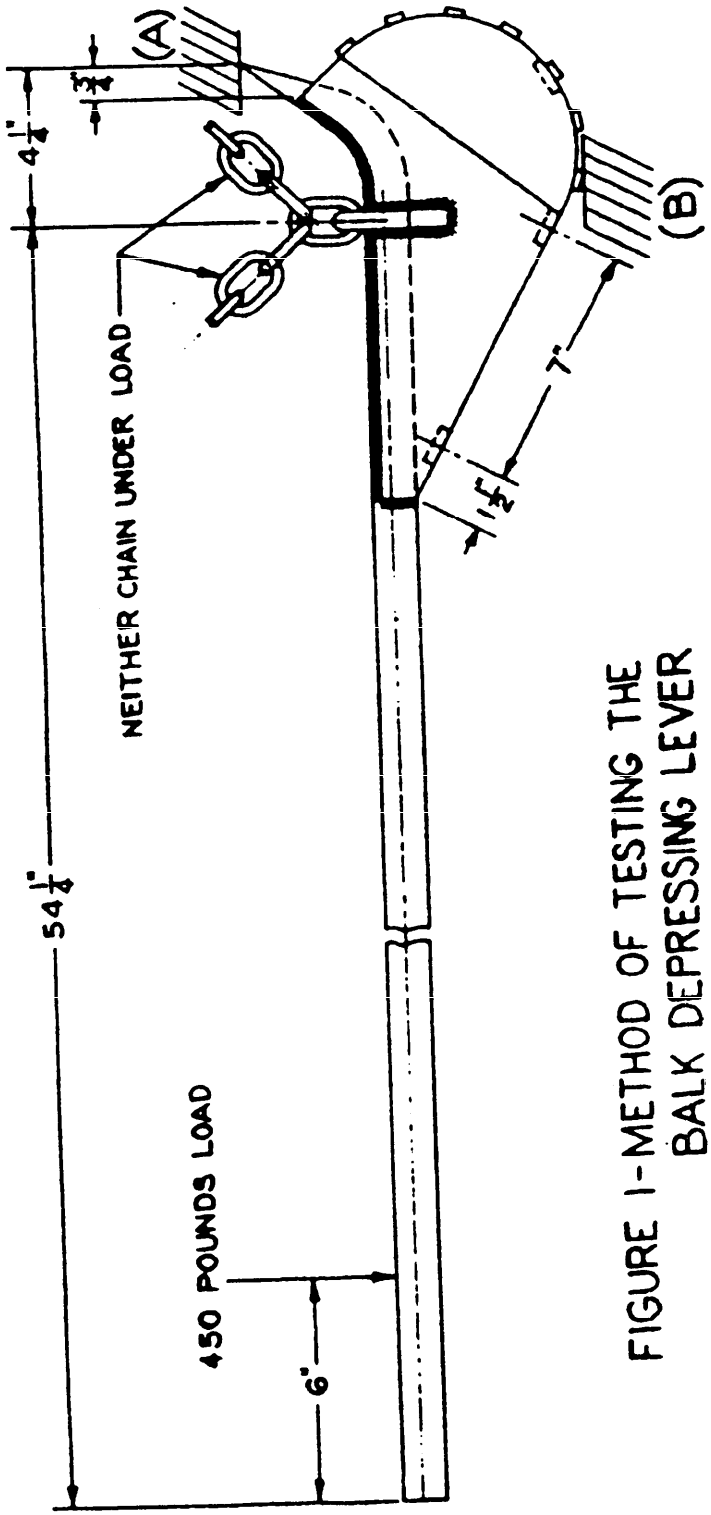


FIGURE 1-METHOD OF TESTING THE BALK DEPRESSING LEVER

NOTE:
THE POINT AND LUG WILL BE UNDER REACTIVE LOADS AT SUPPORTING SURFACES(A) AND (B), RESPECTIVELY.

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-11647E	910626

3. DOCUMENT TITLE
Lever, Balk Depressing: Claw-Bar Type

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (703) 664-5717 (2) AUTOVON: 354-5717
c. ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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